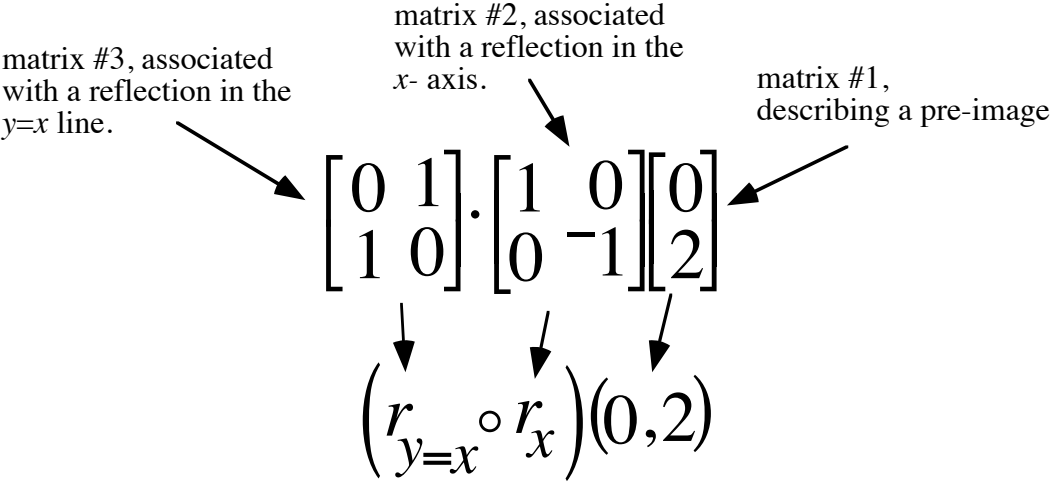
matrix #3, associated with a reflection in the y=x line.
matrix #2, associated with a reflection in the x- axis.
matrix #1, describing a pre-image
$\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\cdot\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}0\\ 2\end{bmatrix}$
$\left(r_{y=x}\circ r_{x}\right)(0,2)$